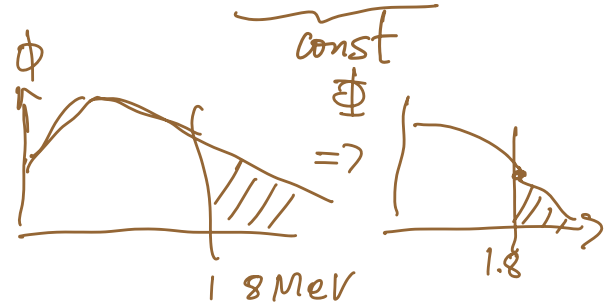


1

In our formalism we are fitting $\Phi(E_\nu)$ for $E_\nu < 1.8 \text{ MeV}$ and NOT $\Delta\Phi = \Phi - \Phi(1.8 \text{ MeV})$. As you can see if we are fitting $\Delta\Phi$, there is a piece missing in R_i

In Eq. (2.3) $\Phi = \int_{E_\nu} d\tilde{E} \phi(\tilde{E}) = \Delta\Phi + \underbrace{\Phi(1.8 \text{ MeV})}_{\text{const}}$

$\phi(E < 1.8 \text{ MeV}) \stackrel{?}{=} \sum \phi_s \delta(E - E_s)$
Do not understand this well



Eq. 4.6 integration by parts

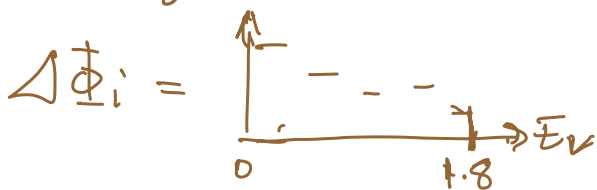
$$\frac{dR}{dE'} = \int_0^{1.8 \text{ MeV}} dE_\nu \frac{\partial \mathcal{H}(E_\nu, E')}{\partial E'} \phi(E_\nu) + \frac{dR}{dE'}(\text{from } E_\nu > 1.8 \text{ MeV})$$

Integration by parts in Eq. 4.7

$$\begin{aligned} \frac{dR}{dE'} &= - \left[\underbrace{\Phi(E_\nu)}_{=0} \frac{\partial \mathcal{H}(E_\nu, E')}{\partial E'} \right]_{E_\nu=0}^{E_\nu=\infty} + \int_0^\infty dE_\nu \Phi(E_\nu) \frac{\partial}{\partial E_\nu} \left[\frac{\partial \mathcal{H}(E_\nu, E')}{\partial E'} \right] \\ &= \int_0^\infty dE_\nu \Phi(E_\nu) \frac{\partial \mathcal{Q}(E_\nu, E')}{\partial E'} \quad (*) \\ &= \int_0^\infty dE_\nu \Delta\Phi_\nu \frac{\partial \mathcal{Q}(E_\nu, E')}{\partial E'} + \Phi(1.8 \text{ MeV}) \int_0^\infty dE_\nu \frac{\partial \mathcal{Q}(E_\nu, E')}{\partial E'} \end{aligned}$$

$$R_i = \int_{E'_i}^{E'_{i+1}} dE' \frac{dR}{dE'} \Rightarrow R_i(E_\nu) = \int_{E'_i}^{E'_{i+1}} dE' \frac{\partial \mathcal{Q}(E_\nu, E')}{\partial E'}$$

$$R_i = \int_0^\infty dE_\nu \Delta\Phi_\nu \mathcal{Q}_i + \Phi(1.8 \text{ MeV}) \int_0^\infty dE_\nu \mathcal{Q}_i(E_\nu)$$



now ansatz for only $E_\nu < 1.8 \text{ MeV}$
Is this what we are doing?

$$R_i = \sum_j \Delta \Phi_j R_{ij} + \int_{E_\nu = 1.8 \text{ MeV}}^{\infty} dE_\nu \left[\Delta \Phi_\nu + \Phi_\nu (1.8 \text{ MeV}) \right] R_i(E_\nu) \quad [2]$$

this is Φ_ν

$$+ 1.8 \text{ MeV} \int_0^{1.8} dE_\nu R_i(E_\nu)$$

$$R_i = \sum_j \Delta \Phi_j R_{ij} + \int_{E_\nu = 1.8 \text{ MeV}}^{\infty} dE_\nu \Phi_\nu R_i(E_\nu) +$$

If we are fitting $\Delta \Phi$ there

would be an
extrapiece in R_i

but this is h_i

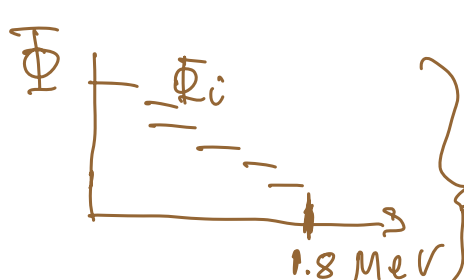
$$\Rightarrow + 1.8 \text{ MeV} \int_0^{1.8} dE_\nu R_i(E_\nu)$$

Extra piece??

Start again from $\textcircled{*}$

$$\frac{dR}{dE'} = \int_0^{\infty} dE_\nu \Phi(E_\nu) \frac{\partial \mathcal{R}}{\partial E'}(E_\nu, E') \quad \textcircled{*}$$

$$\Rightarrow R_i = \int_{E'_i}^{E'_{i+1}} dE' \frac{dR}{dE'} = \int_0^{\infty} dE_\nu \Phi(E_\nu) \mathcal{R}_i(E_\nu)$$



$$R_i = \int_0^{1.8 \text{ MeV}} dE_\nu \Phi(E_\nu) \mathcal{R}_i(E_\nu) + \int_{1.8}^{\infty} dE_\nu \Phi(E_\nu) \mathcal{R}_i$$

h_i

$$R_i = \sum_j \Phi_j R_{ij} + h_i$$

ansatz only to 1.8 MeV

Since this is the expression we used

We are fitting $\Phi(E_\nu)$ NOT $\Delta \Phi(E_\nu)$